MATH-464. HW # 1.

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Problem 1:

Construct a formal proof of the following claim:

Claim:

The set $P = \{x \in \mathbb{R}^n \mid Ax = b,\ Cx \leq d\}$ is a convex set in $\mathbb{R}^n$, where A, b, C, d are appropriately sized matrices / vectors.

Solution:

Let

(1) $P = \{x \in \mathbb{R}^n \mid Ax = b,\ Cx \leq d\}$ and

let $x, y \in P$ that is

(2) $x \in P \Rightarrow Ax = b,\ Cx \leq d$ also

(3) $y \in P \Rightarrow Ay = b,\ Cy \leq d$.

Also, let

(4) $z = \lambda x + (1-\lambda) y$ for some λ, $0 \leq \lambda \leq 1$.

If we show that

(5) $\left[\begin{array}{l} \text{(i)}\ Az = b \quad \text{and} \quad \text{(ii)}\ Cz \le d, \text{ then} \\ z \in P \Rightarrow P \text{ is convex.} \end{array}\right.$

(i) Let me evaluate what Az is:

$$Az = A(\lambda x + (1-\lambda) y) = \quad \text{by } (4)$$

$$= \lambda \underbrace{Ax}_{\text{by } (2),\ b} + (1-\lambda) \underbrace{Ay}_{\text{by } (3),\ = b} = \quad \text{by } (2) \,\&\, (3)$$

$$= \lambda \cdot b + (1-\lambda) b = b. \quad \Rightarrow$$

(6) $Az = b$. first requirement (i) holds. ✓

Now, let's try for the second requirement (ii).

(ii) Let me evaluate what Cz is:

$$Cz = C(\lambda x + (1-\lambda) y) = \quad \text{by } (4)$$

$$= \lambda \underbrace{Cx}_{\le d} + (1-\lambda) \underbrace{Cy}_{\le d} \quad \le \quad \text{by } (2) \,\&\, (3) \text{ resp-ly}$$

$$\le \lambda d + (1-\lambda) \cdot d = \cancel{\lambda d} + d - \cancel{\lambda d} = d. \Rightarrow$$

(7) $Cz \leq d \Rightarrow$ second requirment (ii) holds ✓

by (6) & (7), $z \in P \Rightarrow$

P is convex, as desired.

$\blacksquare$